ENGO 545 – Fall 2026
Hydrographic Surveying
Laboratory exercise #3
Plate check calibration for a single-beam echo sounder (SBES)

Deadline: Mon, Nov 2, 2026 (11:59 PM)

Learning objectives:

After completing this laboratory exercise students should be able to:

- Perform a plate check calibration for a single-beam echo sounder (SBES)
- Program a linear parametric least-squares adjustment in MATLAB, Python, or C/C++
- Conduct any relevant statistical tests and/or generate plots of the results in MATLAB or Python
- Estimate the average speed of sound in water and the draft of the transducer
- Analyse the results from a plate check calibration adjustment
- Write a technical report

Background information:

Several approaches can be employed to determine water depth. Single beam acoustic depth sounding is probably the most widely used depth measurement technique. Acoustic depth measurement systems measure the elapsed time that an acoustic pulse takes to travel from the transducer to the waterway bottom and back. As such, if the velocity of sound propagation in the water column is known, the depth can be computed using following [1]:

$$d = \left(\frac{v \cdot t}{2}\right) + k + d_r$$

where:

d = corrected depth from (reference) water surface
v = average velocity of sound in the water column
t = measured elapsed time from transducer to bottom and back to transducer
k = system index constant/zero error
d_r = distance from (reference) water surface to transducer (a.k.a. draft)

The parameters v, t, and d_r cannot be perfectly determined during the echo sounding process, and k must be determined from periodic calibration of the equipment. The elapsed time, t, is dependent on the reflectivity of the bottom and related signal processing methods used to determine a valid return. The draft, d_r, of the vessel will vary depending on the type of vessel, and the quantity of fuel, water, supplies, etc. on board.

Calibration of acoustic sounding equipment is necessary to ensure good quality soundings are obtained. A common calibration method is the bar or plate check method; see Figure 1 [2]. The bar, or plate, is attached to one or more lines that are marked at known intervals. The plate is suspended beneath the echo sounder at each interval. Any difference between the depth recorded and the known depth represents the correction that should be made to any subsequent soundings at that depth. The calibration should be carried out over the range of expected depths for each survey, as sound velocity may vary throughout the water column. In practice, this test should be carried out at the beginning and end of each day.

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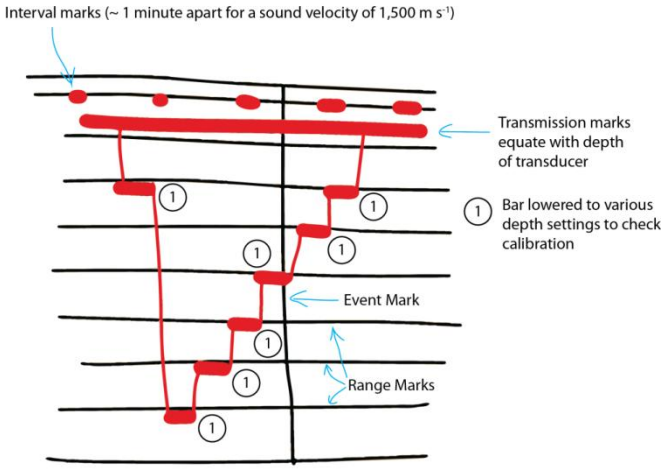


***Figure 1*: A typical bar/plate check with echo sounder record (*after* [2])**

Tasks:

You are asked to perform the following:

(a) Compute the total RMSE between the "raw" sonar and the sounding lead observations based on your field notes from the SonarMite calibration data collection campaign in the deep end of the University of Calgary swimming pool;
(b) **Bonus:** Using your hand-written field notes as a guide, clean up the data collected using the SonarMite+ App, and come up with an average distance value for each of the observed bar/plate intervals;
(c) Using a least squares adjustment (in MATLAB, Python, or C++), estimate the vertical offset and scale factor between the plate depths and the observed sonar distances along with their variance-covariance matrix;
(d) Transform the "raw" sonar observations to corrected depths by applying the estimated parameters in c); What is the RMSE now? Did it improve? Why or why not?
(e) Using the estimated scale factor determine the average velocity of sound in this particular water body along with its ± 95% confidence interval; How does it compare to the theoretically expected value for velocity of sound in water given the observed temperature and depth? Are the two values significantly different? **Hint:** use one of the given empirical models for the speed of sound in water;
(f) Is it safe to assume that the velocity of sound in this particular water body is constant? You may verify this by analyzing the residuals from your least squares adjustment;
(g) How does the estimated vertical offset value compare to the draft measured in the field? Are the two values significantly different? What do you think the system index constant/zero error is? How differently would you have conducted the calibration data collection in order to be able to estimate the system index constant/zero error?

Deliverables:

Check the Additional Course Information document on D2L for the required deliverables on a laboratory exercise.

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References:

[1] US Army Corps of Engineers, “Chapter 4: Single Beam Acoustic Depth Measurement Techniques,” in *Engineering and Design - Hydrographic Surveying (EM 1110-2-1003)*, Washington D. C.: Department of the Army, US Army Corp of Engineers, 2013.

[2] A. E. Ingham, *Hydrography for the surveyor and Engineer*, 3rd Ed. London: Blackwell Scientific Publications, 1992.

[3] H. Medwin, “Speed of sound in water: A simple equation for realistic parameters,” *The Journal of the Acoustical Society of America*, vol. 58, no. 6, pp. 1318–1319, 1975.

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